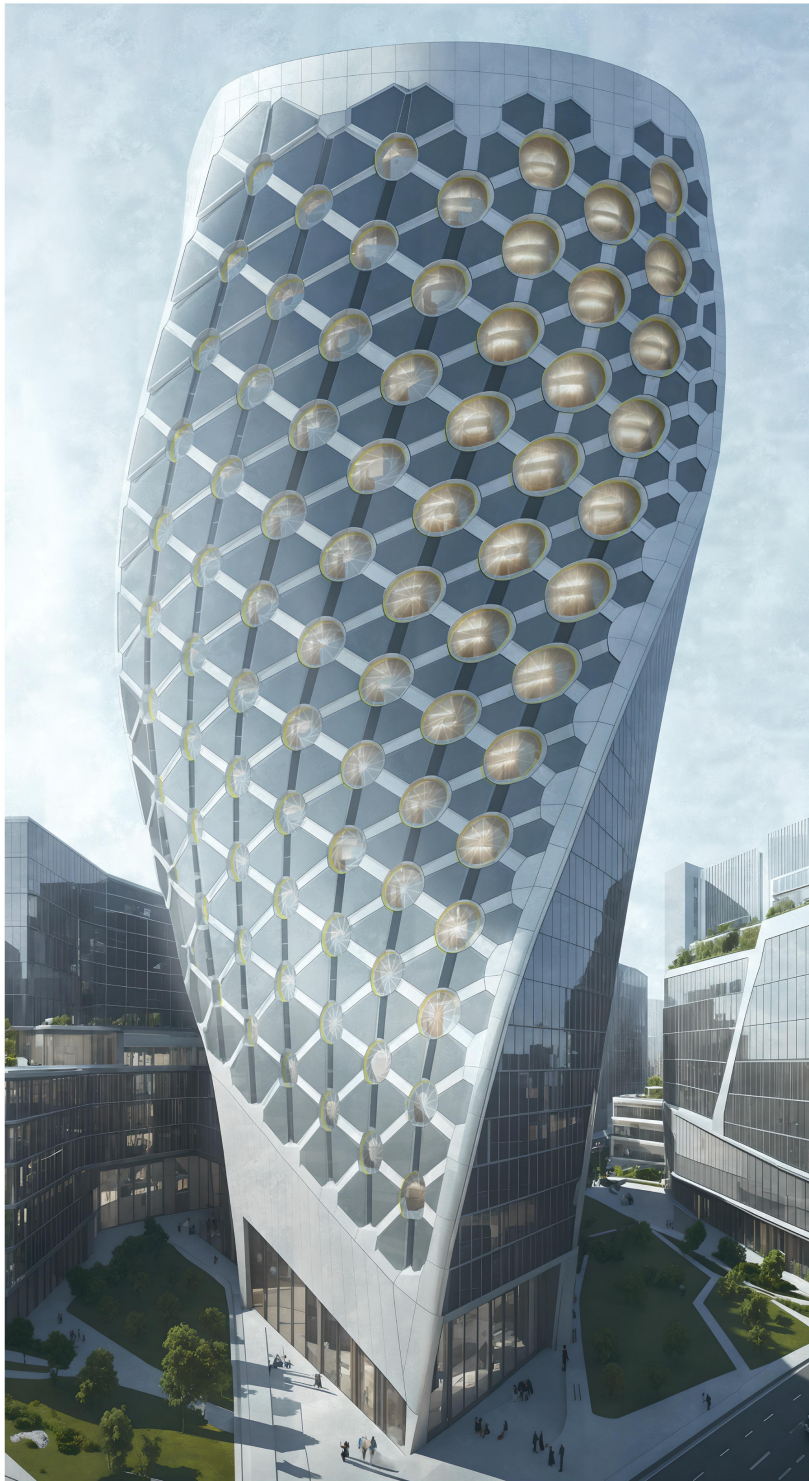




1.1. Concept



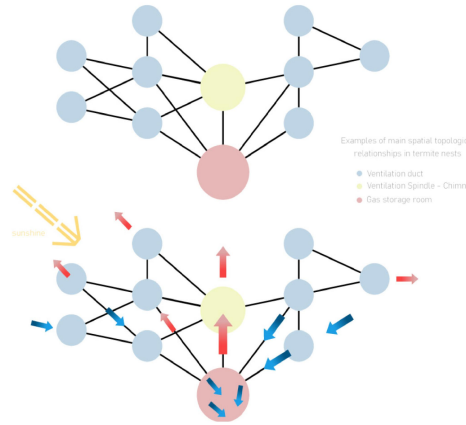
Termite mound



Inner room

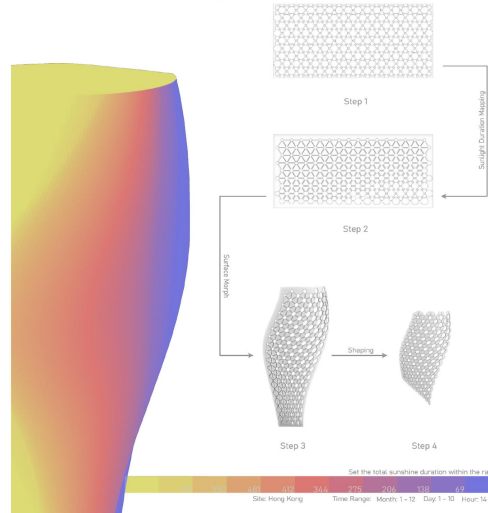
Termite mounds use a clever cooling technique. During the day, hot air rises and exits through the top vents, while cool air is drawn in through the lower vents. As the air passes through the moist soil and mound walls, it cools down through evaporation. This system maintains a stable and comfortable temperature inside the mound, regardless of external conditions.

1.2. Principle research



2.6. Ladybug Sunlight Analysis Optimization

Example:

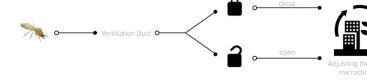


Other Work

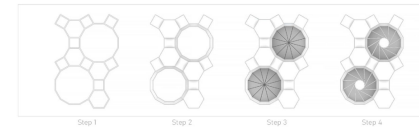
可呼吸的建筑表皮 BREATHABLE FACADE Nature-Inspired Solutions for Sustainable Design

Description: High-rise buildings face two challenges: wind load for stability and natural ventilation. These contradict, as wind loads resist energy while ventilation needs costly, energy-draining air conditioning. To resolve this, we draw inspiration from nature, particularly termite mounds, applying bionic design to create ecological, intelligent facades for sustainable, low-carbon buildings in the digital age.

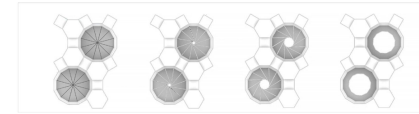
2.1. Prototype reference



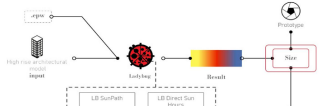
2.2. Prototype



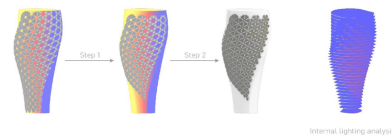
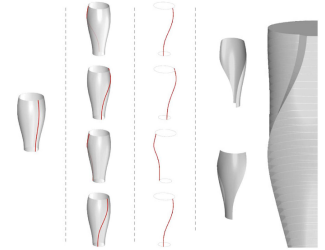
2.3. Running status demonstration



2.4. Deep Conception - Intelligent Facade



2.5. Mass generation of high rise building



2.7. Learning the principles of ventilation in termite mounds

